Math 3
Concept IV + V Review WS

Name: _____
Date: _____ Per: _____

Directions: Show work for all problems involving calculations. All answers must be in simplest form unless otherwise specified.

Simplify each expression. Express your answer in exponential form using only positive exponents.

1) $x^{25} \cdot x^{7}$

x^{25+7}

x^{32}

2) $\frac{5^{8}}{5^{12}}$

5^{8-12}

$5^{-4} \rightarrow \frac{1}{5^{4}}$

3) 8675309^{0}

1

4) $\frac{1}{3^{-4}}$

3^{4} or 81

5) $\left(x^{6}\right)^{2}$

$x^{6 \cdot 2}$

x^{12}

6) $\frac{2 \cdot 7^{5}}{2^{8} \cdot 7^{-11}}$

$\frac{2}{2^{8}} \cdot \frac{7^{5}}{7^{-11}}$

$2^{1-8} \cdot 7^{5-(-11)}$

$2^{-7} \cdot 7^{16} \rightarrow \frac{7^{16}}{2^{7}}$

7) $\frac{5^{-4}}{10^{9}}$

$\frac{1}{5^{4} \cdot 10^{9}}$

8) $\left(8x^{4}y\right)^{3}$

$8^{3}x^{4 \cdot 3}y^{3}$

$512x^{12}y^{3}$

9) $\frac{8x^{4}y^{8}}{24x^{-3}y^{9}}$

$\frac{8}{24} \cdot \frac{x^{4}}{x^{-3}} \cdot \frac{y^{8}}{y^{9}} \rightarrow \frac{1}{3}x^{4-(-3)}y^{8-9}$

$\frac{1}{3}x^{7}y^{-1} \rightarrow \frac{x^{7}}{3y}$

10) $\left(5x^{3}y^{-7}\right)\left(4x^{9}y^{12}\right)$

$(5 \cdot 4)(x^{3} \cdot x^{9})(y^{-7} \cdot y^{12})$

$20x^{3+9}y^{-7+12}$

$20x^{12}y^{5}$

11) Show that if you quadruple 4^{9} you will get 4^{10}

QUADRUPLE MEANS 4 TIMES GREATER

$4 \cdot 4^{9} = 4^{1+9}$

$= 4^{10}$ yay!

For problems 12 and 13, solve for x.

12) $3^{x} \cdot 3^{9} = 3^{900}$

$3^{x+9} = 3^{900}$

$x+9 = 900$

$-9 \quad -9$

$x = 891$

13) $\left(10^{x}\right)^{4} = 10^{30}$

$10^{4x} = 10^{30}$

$\frac{4x}{4} = \frac{30}{4}$

$x = \frac{15}{2}$ or $7\frac{1}{2}$

Want to hear a joke about paper?

Write the given number in scientific notation.

14) 0.000831

8.31×10^{-4}

15) 90,900

9.09×10^{4}

Write the given number in standard form

16) 1.83×10^{4}

18300

17) 9.24×10^{-5}

0.0000924

For problems 18 – 22, evaluate each expression. If necessary, rewrite your answer in scientific notation.

18) $(1.3 \times 10^{4})(2 \times 10^{-8})$

$(1.3 \cdot 2) \times (10^{4} \cdot 10^{-8})$

$2.6 \times 10^{4+-8}$

2.6×10^{-4}

19) $2.3 \times 10^{8} + 4.3 \times 10^{10}$

$.023 \times 10^{10} + 4.3 \times 10^{10}$

$(0.023 + 4.3) \times 10^{10}$

4.323×10^{10}

20) $(3 \times 10^{1000}) - (1.7 \times 10^{999})$

$3 \times 10^{1000} - 0.17 \times 10^{1000}$

$(3 - 0.17) \times 10^{1000}$

2.83×10^{1000}

21) $(1.3 \times 10^{-6}) + 0.00019$

$1.3 \times 10^{-6} + 1.9 \times 10^{-4}$

$0.013 \times 10^{-4} + 1.9 \times 10^{-4}$

$(0.013 + 1.9) \times 10^{-4}$

1.913×10^{-4}

22) $\frac{1.2 \times 10^{-8}}{3 \times 10^{12}}$

$\frac{1.2}{3} \times \frac{10^{-8}}{10^{12}}$

$0.4 \times 10^{-8-12}$

0.4×10^{-20}

4×10^{-21}

Please keep in mind that while this is a good sample of what we have done for Concepts IV and V, it does not represent everything. Please review your CCs 5 - 7 as well.